

Homework (Jalapeno)

- Q1.** In a box-and-whisker plot, what does the line inside the box represent?
(A) Q_2 (median)
(B) Q_1 (first quartile)
(C) Q_3 (third quartile)
(D) Interquartile range
(E) Range
- Q2.** The Rosetta Stone has 15 ancient Greek inscriptions with the following number of lines: 20, 25, 18, 30, 22, 28, 19, 24, 26, 21, 27, 23, 29, 31, 32. What is the lower quartile (Q_1)?
(A) 22
(B) 23
(C) 24
(D) 25
(E) 26
- Q3.** An archaeologist excavated 12 ancient artifacts, with weights (in grams) of: 150, 120, 180, 200, 160, 140, 170, 190, 130, 110, 100, 210. Which weight is an outlier?
(A) 100
(B) 110
(C) 120
(D) 130
(E) 210
- Q4.** What is the interquartile range (IQR) of the following dataset: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q5.** The Great Pyramid of Giza has 35 layers of stone, each with a different height. If the heights are: 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40, 42, 45, 48, 50, 52, 55, 58, 60, what is the upper quartile (Q_3)?
(A) 60
(B) 62
(C) 65
(D) 68
(E) 70
- Q6.** A historian recorded the number of battles won by 10 ancient generals: 5, 8, 12, 15, 18, 20, 22, 25, 28, 30. What is the range of the data?
(A) 20
(B) 22
(C) 25
(D) 28
(E) $30-5=25$
- Q7.** An ancient city has 8 districts with the following population sizes (in thousands): 20, 25, 30, 35, 40, 45, 50, 100. What is the median population size?
(A) 30
(B) 32.5
(C) 35
(D) 37.5
(E) 40
- Q8.** The Terracotta Army has 8000 soldiers, each with a unique height (in cm): 160, 170, 180, 190, 200, 210, 220, 230, 240, 250. If the first quartile (Q_1) is 180, what is the third quartile (Q_3)?
(A) 200
(B) 210
(C) 220
(D) 230
(E) 240

Open Ended Questions (FRQ)

- Q9.** Create a box-and-whisker plot for the following dataset: 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40. Identify the median, first quartile, and third quartile.
- Q10.** A museum has 15 artifacts from ancient civilizations, with the following weights (in grams): 100, 120, 150, 180, 200, 220, 250, 280, 300, 320, 350, 380, 400, 450, 500. Calculate the interquartile range (IQR) and determine if there are any outliers.

Chef's Tips

- Tip 1: Ensure students understand the concept of quartiles and interquartile range.
- Tip 2: Encourage students to use graph paper for box-and-whisker plots.
- Tip 3: Emphasize the importance of accurate calculations and labeling.